Structured Computer Architecture
Chapter 2.11: Processors
Third-Order Systems (3-OS)

An Order-Based Approach

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Based on the textbook

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Outline

Learning Objectives

By the end of this chapter, you will be able to:

- Understand third-order systems (3-OS) and the third feedback loop
- Analyze the structure of elementary processors
- Design RALU (Registers with ALU) components
- Understand CROM (Control ROM) operation
- Explain micro-programming concepts
- Recognize context-free language processing capabilities

Evolution to 3-OS: The Third Loop

- **Previous Systems Review**:
  - **0-OS**: No loops, combinational circuits
  - **1-OS**: One loop, memory capability
  - **2-OS**: Two loops, autonomous behavior
- **3-OS Innovation**: Third feedback loop
  - Can be closed through combinational circuit
  - Can be closed through memory circuit
  - **Focus**: Closed through automaton over automaton
  - Enables processor functionality
- **Key Achievement**:
  - Complex computational behavior
  - Context-free language recognition
  - Elementary processor implementation

Context-Free Languages and Processors

- **Language Hierarchy**:
  - **Type 3 (Regular)**: Recognized by finite automata
  - **Type 2 (Context-Free)**: Require more powerful systems
  - **Type 1 (Context-Sensitive)**: Even more complex
  - **Type 0 (Unrestricted)**: Turing machine equivalent
- **3-OS Capability**:
  - Recognizes context-free languages
  - Processes nested structures
  - Handles recursive patterns
  - Enables programming language parsing
- **Practical Applications**:
  - Compiler design
  - Expression evaluation
  - Nested procedure calls
  - Balanced parentheses checking

Beyond Finite Automata Limitations

- **Finite Automaton Limitation**:
  - Cannot recognize $\{1^n 0^n | n > 0\}$
  - Fixed number of states independent of n
  - No counting capability
- **Counter-Expanded Automaton (CEA)**:
  - Finite automaton + reversible counter
  - Counter provides zero flag to automaton
  - Creates 3-OS system
  - Enables context-free language recognition
- **Enhanced Capabilities**:
  - Counting and comparison operations
  - Stack-like behavior simulation
  - Nested structure processing
  - Context-free grammar parsing

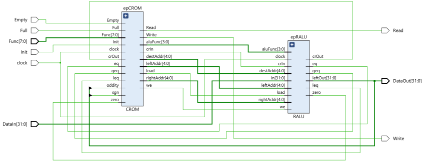
CEA Structure and Operation

Components:

- **Finite Automaton**: Control logic
- **Reversible Counter**: Counting mechanism
- **Zero Flag**: Counter state indicator
- **Command Interface**: Counter control

Counter Operations:

- **NOP**: No operation
- **UP**: Increment counter
- **DOWN**: Decrement counter
- **Zero Flag**: Indicates counter $= 0$



System Integration:

- Both FA and Counter are 2-OS
- Loop connection creates 3-OS
- Enhanced computational power

Elementary Processor Overview

- **Definition**: Simplest version of a processor
  - Implements 3-OS principles
  - Demonstrates processor fundamentals
  - Educational and practical value
- **Main Components**:
  - **RALU**: Registers with Arithmetic & Logic Unit
  - **CROM**: Control ROM (micro-programmed control)
  - **Interface**: Data input/output and control signals
- **Design Philosophy**:
  - Simple functional automaton (RALU)
  - Complex control automaton (CROM)
  - Clear separation of concerns
  - Micro-programmable architecture

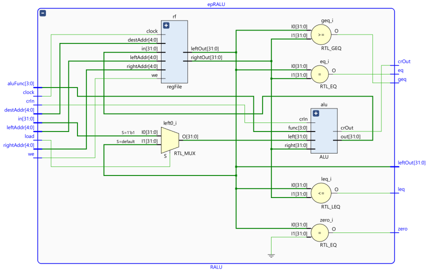
RALU: Registers with ALU

Structure:

- **Register File**: Multi-port storage
- **ALU**: Arithmetic and logic operations
- **Flag Generation**: Status indicators
- **Data Paths**: Interconnection network

Flag Outputs:

- **crOut**: Carry output from ALU
- **eq**: Equal (leftOut == rightOut)
- **leq**: Less than or equal
- **geq**: Greater than or equal
- **zero**: Equal to zero
- **sgn**: Sign bit (negative)
- **oddity**: Odd number indicator



Characteristics:

- Simple functional automaton
- Regular, predictable structure
- Low complexity relative to size
- Efficient implementation